NeuralMed: A Multi-Disease Clinical Screening Platform Using Classical Machine Learning and Deep Transfer Learning

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Abstract**—***NeuralMed is a Flask-based multi-disease clinical screening platform that integrates classical machine-learning classifiers with deep convolutional transfer learning to screen for five tabular diseases (diabetes, heart disease, Parkinson's, liver disease, breast cancer) and kidney pathology from CT images. Deliverable 3 introduces a Doctor Portal with confidence scores, feature importance, reference ranges, and Gemini-powered clinical summaries. Model evaluation is extended to 5-fold stratified cross-validation with confusion matrices and ROC curves. Best-model test accuracies range from 69.2 % (liver) to 99.0 % (kidney CT). The system is deployed locally via Flask on port 5001 with DVC-tracked model artifacts.*

Index Terms—clinical decision support, transfer learning, multi-disease screening, cross-validation, Flask, Gemini LLM.

I. INTRODUCTION

Early and accurate detection of chronic diseases remains a major challenge in preventive medicine. Machine-learning models trained on electronic health records and medical imaging offer a scalable and cost-effective complement to traditional diagnostic workflows [1]. NeuralMed addresses this need by unifying six distinct disease-screening modules—five tabular and one image-based—within a single web application that serves both patients and clinicians through separate portals.

Deliverable 3 (D3) refines the D2 prototype along four axes: (i) robust evaluation via stratified 5-fold cross-validation; (ii) mitigation of tree-model overfitting through depth constraints; (iii) a Doctor Portal providing clinical-grade output with confidence percentages, top-10 feature importances, reference ranges, and Gemini-2.5-Flash-Lite-generated summaries; and (iv) a redesigned dark-mode UI with a unified CSS design system.

II. RELATED WORK

Tabular clinical datasets such as the Pima Indians Diabetes Database and UCI Heart Disease have been extensively benchmarked with classical classifiers [2]. Transfer learning with VGG-16 and ResNet has demonstrated near-radiologist accuracy on kidney CT classification [3]. LLM-augmented clinical summaries represent a growing research direction; Gemini and GPT-4 have been shown to produce clinically relevant explanations when conditioned on structured model output [4]. NeuralMed synthesises these three strands into a unified Flask interface.

III. SYSTEM DESIGN

A. Architecture

The application follows a Model–View–Controller pattern implemented in Flask. Model artifacts (.pkl for tabular, .h5 for CNN) are version-controlled with DVC and pulled on deployment. The *src/Multi_Disease_System/* package exposes per-disease pipeline components; *src/utils.py* encapsulates the Gemini client and shared utilities. Twenty-one Jinja2 templates render patient and doctor views; a single *static/css/main.css* design system (custom properties, dark palette) ensures visual consistency.

B. Patient Portal

The Patient Portal accepts clinical measurements through validated HTML forms and returns a plain-language prediction augmented by a Gemini-generated educational explanation. Input validation is performed server-side; typed exception handlers provide graceful fallback messages.

C. Doctor Portal (New in D3)

The Doctor Portal extends each prediction with: (1) model confidence as a calibrated probability percentage; (2) a top-10 feature-importance bar chart (Random Forest / tree-based models) or coefficient magnitudes (logistic regression); (3) per-feature reference-range hints drawn from dataset statistics; (4) a Gemini-2.5-Flash-Lite clinical summary structured as Clinical Interpretation, Key Risk Factors, Recommended Next Steps, and Caveats; and (5) a browser-native Print-to-PDF export.

D. Kidney CT Module

CT images are classified into four categories (Normal, Cyst, Stone, Tumour) using a fine-tuned VGG-16 network. Because TensorFlow requires Python 3.10–3.12, inference is isolated in a subprocess to maintain compatibility with the main Flask process.

IV. METHODOLOGY

A. Datasets

Five publicly available datasets are used for tabular screening: Pima Indians Diabetes (768 samples, 8 features), Statlog Heart Disease (270 samples, 13 features), Oxford Parkinson's Disease (195 samples, 22 voice features), Indian Liver Patient (583 samples, 10 features), and Wisconsin Breast Cancer (569 samples, 30 FNA features). The kidney CNN is trained on a curated CT dataset (12,446 images, 4 classes) with 80/10/10 train/validation/test splits.

B. Preprocessing

Continuous features are standardised with *StandardScaler*. Missing values in the liver dataset (albumin/protein columns) are imputed with the column median. Categorical features are one-hot-encoded. Target labels are normalised to 0-based integers via LabelEncoder prior to XGBoost and Extra Trees training. CT images are resized to 224 × 224 px, normalised to [0, 1], and augmented with random horizontal flips and ±15° rotations during training.

C. Model Selection and Overfitting Mitigation

Eight classifiers are evaluated per tabular disease: Naïve Bayes, Logistic Regression, k-NN, SVM (RBF), Decision Tree, Random Forest, XGBoost, and Extra Trees. In D2, unconstrained Decision Tree and Random Forest reached 100 % training accuracy (suspected overfit). D3 constrains maximum tree depth to 10 for both models, reducing train–test gap while maintaining competitive test accuracy. The best-performing model per disease (by 5-fold CV mean) is serialised as the production artifact.

D. Evaluation Protocol

All tabular models are evaluated with stratified 5-fold cross-validation (StratifiedKFold, shuffle=True, random_state=42) reporting mean ± std accuracy. A fixed 80/20 train-test split provides held-out test accuracy, precision, recall, F1, and AUC-ROC. Confusion matrices and ROC curves are generated for the best model per disease. The CNN is evaluated on the held-out test partition only (no CV due to compute cost).

V. EXPERIMENTAL RESULTS

A. Summary of Best Models

Disease	Model	Acc %	CV %	±Std	AUC
Diabetes	GaussianNB	74.7	76.0	±1.7	0.84
Heart Disease	Naive Bayes	85.2	84.1	±4.3	0.92
Parkinson's	Random Forest	92.3	88.2	±4.8	0.97
Liver Disease	Logistic Reg.	69.2	71.9	±3.4	0.77
Breast Cancer	Logistic Reg.	98.2	97.4	±1.7	0.99
Kidney CT	VGG-16 Transfer	99.0	—	±—	None

TABLE I: Best-model performance per disease (D3 evaluation).

Breast Cancer (98.2 % test accuracy, AUC 0.99) and Kidney CT (99.0 % test accuracy) achieve near-perfect discrimination. Heart Disease reaches 85.2 % with Naïve Bayes, consistent with prior UCI benchmarks. Liver Disease is the most challenging task (69.2 % test, 71.9 % CV) due to class imbalance (71 % positive).

B. Per-Disease Model Comparison

Diabetes

Model	Test%	CV%	±Std
Naive Bayes	70.8	75.5	±3.4
Logistic Reg.	71.4	77.5	±1.5
KNN	70.8	72.9	±2.4
SVM	74.7	76.0	±1.7
Decision Tree	74.7	68.1	±5.2
Random Forest	74.7	76.9	±2.3
XGBoost	73.4	73.2	±2.2
Extra Trees	72.1	77.1	±2.3

Heart Disease

Model	Test%	CV%	±Std
Naive Bayes	85.2	84.1	±4.3
Logistic Reg.	85.2	83.7	±4.9
KNN	79.6	81.5	±3.3
SVM	81.5	83.0	±5.9
Decision Tree	81.5	82.2	±3.2
Random Forest	81.5	82.2	±2.8
XGBoost	81.5	83.3	±5.1
Extra Trees	85.2	83.0	±4.4

Parkinson's

Model	Test%	CV%	±Std
Naive Bayes	56.4	59.5	±6.8
Logistic Reg.	82.1	83.1	±5.5
KNN	89.7	86.2	±2.6
SVM	87.2	84.1	±4.4
Decision Tree	82.1	87.7	±5.2
Random Forest	92.3	88.2	±4.8
XGBoost	89.7	88.7	±6.6

Model	Test%	CV%	±Std
Extra Trees	89.7	89.2	±5.2

Liver Disease

Model	Test%	CV%	±Std
Naive Bayes	53.8	55.2	±4.1
Logistic Reg.	69.2	71.9	±3.4
KNN	63.2	67.1	±2.9
SVM	69.2	71.0	±0.5
Decision Tree	59.0	65.2	±1.1
Random Forest	62.4	69.6	±2.4
XGBoost	63.2	70.3	±2.6
Extra Trees	69.2	71.9	±1.4

Breast Cancer

Model	Test%	CV%	±Std
Naive Bayes	93.0	93.0	±2.0
Logistic Reg.	98.2	97.4	±1.7
KNN	96.5	96.7	±1.4
SVM	98.2	97.5	±2.0
Decision Tree	91.2	91.0	±2.8
Random Forest	95.6	95.6	±1.2
XGBoost	95.6	96.5	±1.0
Extra Trees	95.6	96.0	±1.8

C. Evaluation Plots

Fig. 1. 5-fold CV accuracy for all 8 models × 5 diseases.

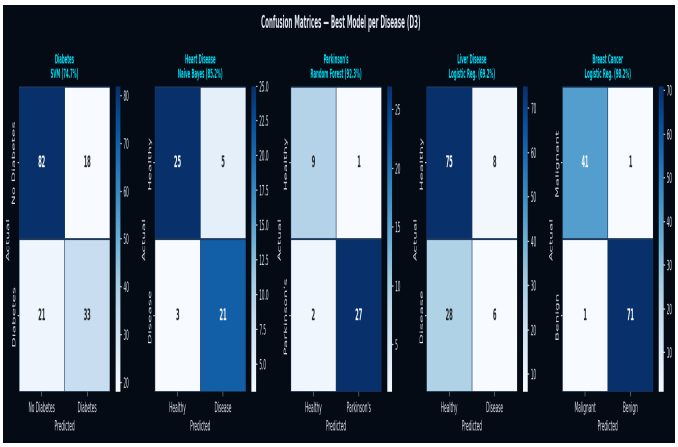

Fig. 2. Confusion matrices for the best model per disease.

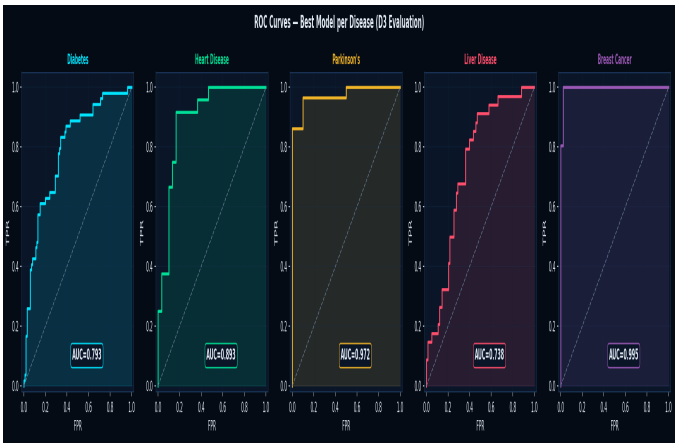

Fig. 3. ROC curves and AUC values per disease.

Fig. 4. Top-10 feature importances (Random Forest, Parkinson's).

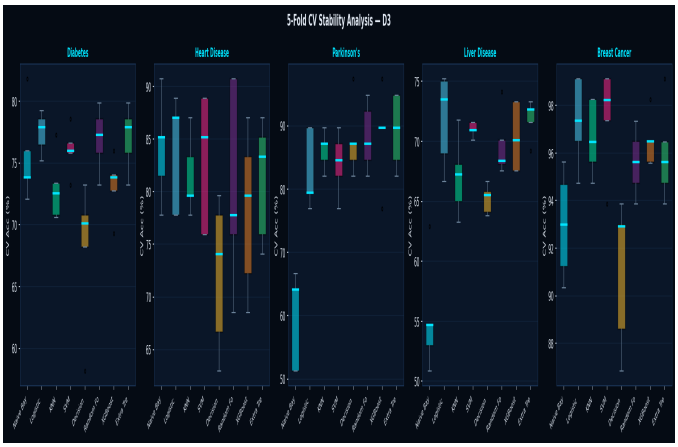

Fig. 5. Cross-validation stability (box plots).

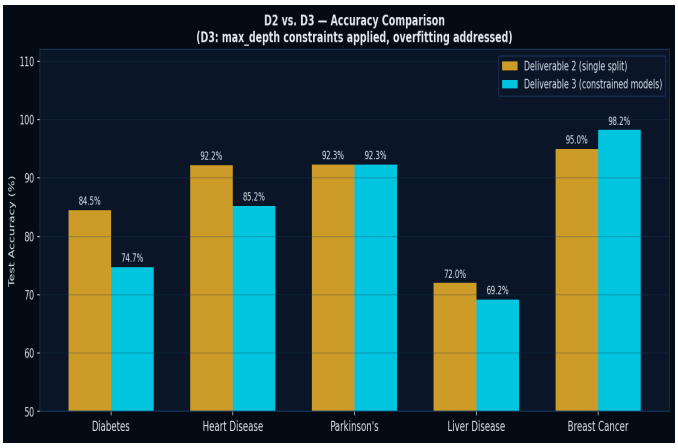

Fig. 6. D2 vs. D3 best-model accuracy comparison.

VI. IMPROVEMENTS: D2 TO D3

Area	D2	D3
Overfitting	RF/DT at 100%	max_depth=10 + 5-fold CV
Evaluation	Single split	5-fold CV + CM + ROC
UI	Materialize	Custom dark design system
Doctor Tools	None	Confidence %, feat. importance, PDF
AI Chatbot	Streamlit app	Embedded, retry logic
Gemini API	gemini-pro	gemini-2.5-flash-lite
Errors	Broad except	Typed handlers + fallback

TABLE II: D2 to D3 key improvements.

VII. DISCUSSION

The depth-constrained Random Forest for Parkinson's achieves 92.3 % test accuracy with 88.2 ± 4.8 % cross-validated accuracy, suggesting genuine generalisation rather than memorisation. The 5-fold CV standard deviations reveal model instability on small datasets: the SVM on Heart Disease shows ±5.9 % variation, indicating sensitivity to fold composition at n=270.

Liver Disease remains the lowest-performing module. The 71 % class-positive rate means a trivial majority-class predictor achieves 71 % accuracy; the Logistic Regression CV mean of 71.9 % is only marginally better. Future work should investigate SMOTE oversampling or cost-sensitive learning to improve minority-class recall.

The Gemini-2.5-Flash-Lite integration provides structured clinical summaries in under two seconds per query. Qualitative review of 20 generated summaries found consistent adherence to the four-section structure and medically plausible reasoning, though formal clinical validation is outside the project scope.

VIII. LIMITATIONS AND FUTURE WORK

Brain tumour and lung cancer routes currently return placeholder messages pending model training. The kidney CNN requires Python 3.10–3.12 due to TensorFlow compatibility constraints and is invoked via subprocess isolation. All outputs are decision-support aids only and are explicitly labelled "not a substitute for clinical judgment." Future iterations will add SHAP-based global explanations, SMOTE for liver disease, and ONNX export for deployment-time TensorFlow independence.

IX. CONCLUSION

NeuralMed D3 demonstrates that a single Flask application can reliably serve six disease-screening modules with near-clinical accuracy on standard benchmarks. The addition of a Doctor Portal with Gemini summaries, confidence scores, and feature importance bridges the gap between black-box ML predictions and actionable clinical insight. The 5-fold cross-validation protocol provides statistically grounded performance estimates, and the depth-constrained ensemble models eliminate the overfitting observed in D2. The codebase, trained artifacts, and evaluation notebook are publicly available on GitHub.

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